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An EMT 2026 Climate Change awareness series on options for Carbon Neutrality



Taken from WG-111, AR6- IPCC (2022)

A safe climate implies a safer future for all mankind. A robust and accelerated global action towards net zero GHG emission is required across every sector in every country. Let us make carbon neutrality a vision for every sector of the economy – EMT, 2026-2030 Climate Action Agenda

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A major shift from a fossil fuel dependent economy to a renewable energy driven system will help to reduce GHG emissions, enhance sustainability and climate resilience. Let us support decarbonization and sustainable energy transition.

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Editorial

The earth's climate system is warming and this is unequivocal. The human influence on the climate system is very clear. The physical science basis of climate change and available evidences have proved that every Green House Gas (GHG) emitted or added to the earth atmosphere will increase the earth's temperature and further accelerate global warming. As humans, we need to ambitiously cut down our CO_2/GHG emissions by making a fundamental departure from business as usual if we must halt global warming and any future irreversible changes on our climate system. We are fast approaching a climate tipping point and the climate time bomb is ticking. This is the time to act in solidarity to halt further global warming. It must be an all-hands-on deck approach across every sector in every country and every economy.

The consequences of global climate inaction are already impacting ecosystems, people, cities and existing infrastructures. Climate change will continue to cause wide spread disruption to humanity and natural systems. The inability to cope with the impacts of climate change will increase vulnerabilities across all regions of the world. Because climate change will continue to be the largest stress multiplier of our time.

Obviously, our decisions and actions today will define and decide our future climate. We need to act with climate justice as a core value. Nations and governments must strengthen their commitment to their NDC- Nationally Determined Contributions to GHG emissions reductions. Robust and ambitious reductions in GHG emissions will be essential to avoid dangerous and irreversible changes in the climate system that could have devastating consequences on our live supporting systems. Humanity must act now to avoid altering our life support systems. We therefore need a change in behaviour, best available technology and finance. Behavioural changes will include but not limited to the following:

- *Human/individual changes in behaviour such as the choices we make, life style and consumption pattern*
- *production pattern, effective reporting of environment, social and governance (ESG) issues by industries and corporations*
- *National and sub-national governments need changes in policies and institutional knowledge and know-how*
- *International bodies must enable equitable transfer of climate finance and technologies for climate change mitigation and adaptation*

Mitigation and inclusive adaptation planning and actions with viable adaptational options and measures such as access to quality health care services, sanitation, fresh water supply, healthy ecosystems, finance and equitable funding will increase the adaptive capacity of people and communities while at the same time reduce vulnerability and increase resilience to climate change. Effective adaptation and mitigation actions will definitely reduce climate change risks. This will be possible if we make today, the right changes and adjustments in human behaviours and avoid unsustainable production and consumption patterns to enable us limit global warming to 2 degrees Celsius and well below 1.5 degrees Celsius. Carbon Neutrality, Net Zero emissions of GHG, efficient energy transition plan, emission cuts/reduction pathways, and scaling up technologies and investment in renewable energy systems will help humanity to achieve a desired goal.

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Developing countries and economies in transition would be most affected by climate change if we delay to act now. We must act to reduce poverty and global inequality in finance distribution, technology and information. This is what makes people vulnerable. Therefore, a one US dollar investment in mitigation and climate change adaptation, financing will save the world and humanity from several US dollars in response to climate change disaster in the future.

We need to learn from available climate change science because learning is vital to understand what has changed, the contexts and drivers to climate change, what will change and future impacts of climate change in addition to best response and intervention options to cope and build required resilience to a changing climate.

To achieve climate resilience, efficient and effective climate solution requires both proactive and SMART climate adaptation and mitigation programme implementation. Hence, we will need:

- *Resources such as knowledge, finance and technology*
- *Governance and institutional arrangement to facilitate coordination*
- *Leadership from top to bottom (national, subnational and international) and across every sector*

Climate change will bring with it additional level of risks such as but not limited to:

- *Risk to people and infrastructures,*
- *Risks to ecosystems and ecosystem services*
- *Risk to fresh water resources, agriculture and food security*

Our agenda should be to pursue a climate resilient development pathway. This is development that integrates measures to adapt to climate change with actions to reduce GHG emissions that provides wider benefits to people and nature. These will require ambitious changes in transportation systems, city infrastructure designs and development, shifts to renewable energy systems (energy transition) - major departure from fossil fuel driven economies, investment into circular economy and sustainable waste management, sustainable and resilient farming and agricultural practices such as changes in planting time, use of different and climate resistant varieties of seeds etc.



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Net-Zero Emission Plan for Green-House Gas Emitting Facilities in Port Harcourt

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Editor's Note & Commentary: The authors analyzed opportunities to achieve net zero emission of greenhouse gases (GHG) across sectors and facilities within sectors with potential to emit GHG's in the selected study area. It is expected that the findings of the authors can support regional governments to improve policies and actions for climate change mitigation and adaptation.

Disclaimer: The data and facts presented here were not subjected to any verification by the ClimateTimes journal team and as such are purely research-based findings by the authors within their study area.

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Abstract
Port Harcourt, Rivers State, Nigeria, is a major industrial and petroleum hub whose activities emit greenhouse gas (GHG). This study developed a facility-level net-zero emissions plan for major GHG-emitting facilities in the Port Harcourt metropolitan-industrial area, thereby supporting the achievement of Nigeria's national determined contribution (NDC), carbon-neutrality ambition and facility-level decarbonization. A cross-sectional, facility-level assessment was conducted across manufacturing, oil and gas, transportation, and solid waste sectors, covering three facilities per sector. Questionnaire were administered to identify GHG-emitting activities, associated GHG types, perceived health and safety impacts, and mitigation opportunities. Descriptive statistics were used to examine facility-level response patterns. Major identified emission-generating activities included fossil-fuel combustion, industrial processes, transportation-related fuel use, oil and gas operational activities, and waste-management processes. Mitigation measures receiving comparatively strong support included energy efficiency, renewable and lower-carbon energy options, operational improvements, carbon-management measures, and sector-specific technological interventions. Mitigation profiles were relatively consistent across transportation and oil and gas facilities, whereas statistically significant facility-level variations occurred in manufacturing and solid waste facilities. The findings informed a five-stage net-zero roadmap comprising baseline GHG accounting and governance, rapid operational abatement, energy-system transformation, deep technology-based decarbonization, and neutralization of residual emissions. The study concludes that sector-wide climate strategies should be complemented by facility-specific action plans because the relevance, priority, and perceived feasibility of mitigation interventions vary among facilities.

Key Words: Net-zero emissions; greenhouse gases; mitigation opportunities; industrial decarbonization; Port Harcourt; Nigeria; facility-level climate action

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Introduction

Climate change has evolved from being viewed primarily as an environmental issue to a broader challenge encompassing economic development, public health, energy security, and sustainable growth. The increasing concentration of greenhouse gases (GHGs) in the atmosphere, particularly carbon dioxide (CO_2), methane (CH_4), and nitrous oxide (N_2O), has intensified global warming, resulting in rising temperatures, altered precipitation patterns, more frequent extreme weather events, sea-level rise, and ecosystem degradation. Human activities, especially industrialization and energy production, remain among the most significant contributors to global GHG emissions (Intergovernmental Panel on Climate Change, 2023).

Climate change is considered a major threat to growth and development in Africa. It is predicted that climate change will in the future adversely affect key socioeconomic sectors as well as the quantity and quality of natural resources within the continent (Patricks, 2020). The challenge is particularly pronounced in developing countries such as Nigeria, where economic growth, poverty reduction, industrial expansion, and energy security must be balanced against climate change mitigation objectives. Unlike developed countries that historically followed carbon-intensive industrialization pathways, Nigeria faces growing pressure to pursue a more sustainable development model while meeting the needs of its rapidly expanding population.

Recognizing this challenge, Nigeria has committed to achieving net-zero greenhouse gas emissions by 2060. To support this objective, the Federal Government introduced the Energy Transition Plan (ETP), which provides a framework for reducing emissions while simultaneously expanding energy access and promoting economic development. The ETP focuses on five key sectors: power, oil and gas, transport, cooking, and industry. Importantly, the plan acknowledges the need to align climate action with national developmental priorities (Federal Republic of Nigeria, 2022). Furthermore, the Climate Change Act of 2021 established a legal framework for coordinating climate actions and supporting the country's decarbonization agenda.

However, national policies alone cannot guarantee emissions reductions. Achieving net-zero emissions requires the translation of national targets into measurable actions at facility level through emissions inventories, operational controls, investment programs, performance targets, and monitoring systems.

According to Patricks (2025) there are options in every sector to reduce GHG emissions either by technology or by behavioural change. This requirement is particularly relevant in Port Harcourt; a major industrial and commercial hub located in the Niger Delta region of Nigeria. The city is surrounded by extensive petroleum exploration, production, processing, transportation, and support-service activities. Industrial operations in the area contribute to GHG emissions through fuel combustion, gas flaring, methane venting, electricity consumption, industrial transportation, and waste management practices. In addition to contributing to climate change, these activities have significant implications for local environmental quality.

Although pollutants such as particulate matter, Sulphur oxides, and nitrogen oxides are not greenhouse gases, their co-occurrence with carbon dioxide and methane highlights the interconnected nature of climate mitigation and environmental management.

The transition to net-zero greenhouse gas emissions increasingly requires climate commitments to be translated from national targets into actions at sector, organizational, and facility levels. Net zero is particularly challenging for industrial and infrastructure-dependent sectors because emissions arise from multiple activities, including combustion, industrial processes, electricity consumption, transportation, fugitive releases, and waste management.

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The IPCC identifies energy efficiency, electrification, low- and zero-GHG energy carriers, material efficiency, circular-economy approaches, and carbon capture among the strategies relevant to industrial decarbonization (Intergovernmental Panel on Climate Change, 2022).

This requirement is particularly relevant to Nigeria. The country's NDC 3.0 provides an economy-wide pathway towards net-zero emissions by 2060 and identifies mitigation measures across fuel combustion, fugitive emissions, industrial processes, transport, and waste. Nigeria's climate objectives consequently require implementation mechanisms capable of converting national targets into operational changes at emitting facilities. (Nigeria NDC, 2025)

Port Harcourt provides an important setting for such investigation because of the concentration of petroleum, manufacturing, transportation, commercial, and waste-management activities in the metropolitan-industrial area. The diversity of these activities means that the pathways through which GHGs are generated are unlikely to be identical across facilities, even when facilities belong to the same sector.
A critical conceptual distinction is therefore required. GHG emission sources refer to activities or processes through which emissions are generated, whereas GHG types refer to the gases emitted. For example, combustion, gas flaring, industrial processes, vehicular fuel use, and waste decomposition are emission-generating activities, whereas carbon dioxide (CO_2), methane (CH_4), and nitrous oxide (N_2O) are greenhouse gases.

A facility-level net-zero plan also requires a consistent approach to GHG accounting. The GHG Protocol Corporate Standard provides a framework for preparing organizational GHG inventories, while its associated guidance addresses purchased electricity and value-chain emissions. However, the questionnaire data used in the present study identified GHG emission activities and mitigation opportunities rather than directly quantifying CO_2-equivalent emissions. Consequently, the present analysis is used to establish net-zero priorities and implementation pathways, while a verified physical GHG inventory is recommended as the first implementation stage. (GHG Protocol, 2024).

The rationale for this approach is important. Two facilities may undertake broadly similar activities but differ in their energy systems, technologies, equipment age, process configuration, management arrangements, and readiness to adopt low-carbon interventions. Thus, sector-wide strategies alone may fail to address facility-specific barriers or opportunities.

Accordingly, this study developed a facility-level net-zero emission plan for GHG-emitting facilities in Port Harcourt based on empirical evidence concerning identified GHG-emitting activities and mitigation opportunities across manufacturing, oil and gas, transportation, and solid waste facilities.
The study objectives sought to:

i. Identify the principal GHG-emitting activities and associated mitigation opportunities among selected manufacturing, oil and gas, transportation, and solid waste facilities in Port Harcourt.
ii. Develop a phased facility-level net-zero emissions framework based on the identified emission priorities, mitigation opportunities, and facility-level variations.

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Methodology

The study adopted a cross-sectional quantitative research design. Primary data were collected using a structured questionnaire administered to respondents across selected manufacturing, oil and gas, transportation, and solid waste facilities in Port Harcourt. The questionnaire was designed to obtain information on identified greenhouse gas (GHG) emission-generating activities, associated GHG types, perceived health and safety impacts, and potential mitigation measures.

The cross-sectional design was considered appropriate because it enabled the collection of data from multiple facilities at a single point in time and facilitated comparisons of GHG-related characteristics and mitigation opportunities across the sampled sectors and facilities.

Following the statistical analysis of all the responses, the findings were used to develop a facility-level net-zero emissions planning framework. The framework was not a research design but an outcome of the study, providing a structured approach for translating the identified emission reduction priorities and mitigation opportunities into practical decarbonisation actions for facilities operating within the Port Harcourt metropolitan-industrial area.

Study Population and Facility Categorization

Four emitting sectors within the Port Harcourt metropolitan-industrial area were investigated. These are:

- manufacturing
- oil and gas
- transportation
- solid waste management

Three facilities designated as A, B, and C., were selected from each sector

Below given number of responses were collected and analyzed for each sector from the sampled facilities

- 403 for Manufacturing
- 407 for Oil and Gas
- 401 for Transportation
- 398 for Solid Waste

Data Collection Instrument

The structured questionnaires contained four major constructs:

1. Sources of GHG (S.GHG) representing identified GHG emission sources/activities;
2. Types of GHG (T.GHG) representing identified types of GHG;
3. Health and Safety Impacts (HIS) for perceived health and safety impacts
4. Mitigation Options (MO.GHG) representing identified GHG mitigation options.

Responses were measured on a five-point Likert scale using questionnaire statements that are sector specific. Consequently, inferential statistical analyses (correlation, ANOVA, Tukey HSD, and chi-square tests) were performed primarily among facilities within the same sector because respondents answered an identical set of sector-specific questionnaire variables. This ensured the validity of the statistical comparisons by maintaining consistency in the variables being analysed.

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Assessment of Net-Zero Priorities
The net-zero planning analysis adopted a priority-based approach.
For each sector, GHG source and mitigation-option were assessed using:

- variable means;
- standard deviations;
- percentage agreement;
- facility-level average scores derived from the combined responses to questionnaire variables measuring the same construct.
- between-facility variations; and
- inferential statistics.

Higher mean and agreement scores were considered indicative of stronger respondent recognition of the emission-generating activity or mitigation opportunity.

It is important to note that these scores represent respondent assessments, not measured tonnes of emissions or quantified technical abatement potential.

Statistical Analysis
Descriptive statistics were used to determine the relative importance of identified emission-generating activities and mitigation options. One-way Analysis of Variance (ANOVA) was used to determine whether the mean composite scores for GHG-emitting activities, GHG types, health and safety impacts, and mitigation options differed significantly among Facilities A, B, and C within each sector. Where significant differences were observed, Tukey's Honestly Significant Difference (HSD) post-hoc test was conducted to identify the specific facilities responsible for the differences. Pearson product-moment correlation analysis was used to examine relationships between GHG-emitting activities, GHG types, health and safety impacts, and mitigation options within each facility. Chi-square tests of independence were used to determine whether the distribution of responses across the five Likert-scale response categories differed significantly among facilities within each sector. Statistical significance was established at $p < .05$.
Composite Mean: This is the average score obtained by combining responses to multiple questionnaire variable measuring the same construct.



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Results and Discussion

Greenhous gas (GHG) Source

Table 1: Summary of GHG Source Scores by Sector

Sector	Number of Source variables	Highest variable Mean	Lowest variable Mean	Overall Pattern
Manufacturing	3	4.08	3.18	Moderate-high recognition
Oil & Gas	5	3.75	3.15	More heterogeneous
Transportation	3	4.07	3.20	Moderate-high recognition
Solid Waste	3	3.95	3.63	Relatively consistent

GHG Emission sources from Manufacturing Sector

Respondents from the Manufacturing sector recognized the principal emission-generating activities represented by S. GHG1, which recorded a mean of 4.08 ± 1.11, with approximately 72.7% agreement.
The remaining GHG-emitting activity variables assessed in the questionnaire recorded mean scores of 3.58 and 3.18, respectively. More importantly from a net-zero planning perspective, the three manufacturing facilities were highly consistent. The facility-level composite means (overall average scores derived from all GHG-source questionnaire variables) ranged from approximately 3.59 to 3.63 and ANOVA confirmed no significant difference:
F (2, 400) = 0.076, p = .927.
This suggests that the GHG emission sources considered in the assessment were perceived similarly across the three manufacturing facilities.

GHG Emission sources from Oil and Gas Sector

Oil and Gas presented a more heterogeneous source profile.
The five questionnaire statements used to assess GHG-emitting activities (S.GHG1-S.GHG5) recorded mean scores ranging from 3.15 to 3.75.
Unlike Manufacturing, GHG-source composite scores differed significantly between Oil and Gas facilities:
F (2, 404) = 5.62, p = .004, η^2 = .027. Facility A recorded the highest composite score (M = 3.73), compared with Facility B (M = 3.41) and Facility C (M = 3.57).

Following the significant ANOVA result, Tukey's Honestly Significant Difference (HSD) post-hoc test was conducted to determine which facilities differed from one another. The analysis showed that Facility A recorded significantly higher GHG-source scores than Facility B (mean difference = 0.319, p = .003). However, Facilities A and C and Facilities B and C did not differ significantly (p > .05). This indicates that the overall variation observed within the Oil and Gas sector was primarily attributable to differences between Facilities A and B rather than differences among all three facilities.

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ANOVA only indicates that a significant difference exists somewhere among the three facilities but does not identify the specific facilities involved. Therefore, Tukey HSD multiple comparison tests were performed to determine which facility pairs differed significantly.

This result suggests that although the Oil and Gas facilities operate within the same sector and may share similar categories of emission-generating activities, the relative importance attributed to those activities differ between facilities.

Therefore, the findings do not imply that one oil and gas facility necessarily possesses completely different GHG sources from another. Instead, they support conducting a facility-specific hotspot inventory before assigning individual abatement priorities.

Nigeria's current Nationally Determined Contribution (NDC) similarly distinguishes mitigation action relating to fuel combustion and fugitive emissions, reflecting the need to target individual emission pathways. (Nigeria NDC, 2025)

GHG Emission sources from Transportation Sector

Transportation demonstrated strong consistency.

Among the identified GHG-emitting activities, the activity represented by S.GHG1 recorded the highest mean score (4.07 ± 1.13), while the remaining variables averaged 3.56 and 3.20. Facilities A, B, and C showed remarkably similar variable scores. The source composite means were approximately:

- Facility A: 3.62
- Facility B: 3.61
- Facility C: 3.59

ANOVA confirmed that these differences were not statistically significant: F (2, 398) = .074, p = .928.

The similarity implies that a common initial decarbonization framework may be particularly feasible across the sampled transportation facilities. Globally, transport decarbonization requires combinations of demand reduction, improved efficiency, electrification, alternative fuels, and infrastructure transformation rather than dependence on one technological intervention. (Intergovernmental Panel on Climate Change, 2022)

GHG Emission sources from Solid Waste Sector

Solid Waste source variable recorded means of:

- 3.95;
- 3.76; and
- 3.63.

Corresponding agreement percentages were approximately 69.8%, 67.1%, and 60.8%.

Despite some variation in response distributions, mean composite source scores did not differ significantly between Facilities A, B, and C:

F (2, 395) = .509, p = .601.

The findings suggest that the facilities broadly recognize similar waste-related emission challenges, although previous chi-square analysis demonstrated that the distribution of individual Likert responses differs somewhat between facilities.

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GHG Mitigation Opportunities

The findings on mitigation opportunities are particularly important for developing the proposed net-zero pathway.

Table 2: Questionnaire-Derived Mitigation Opportunity Profile

Sector	Mitigation Variable Mean Range	Highest Agreement	Facility-Level Difference
Manufacturing	3.77-4.01	85.1%	Significant
Oil & Gas	3.12-3.88	68.1%	Not significant
Transportation	3.75-4.24	91.3%	Not significant
Solid Waste	3.33-3.99	74.6%	Significant

Manufacturing Mitigation Opportunities

Manufacturing demonstrated strong support for mitigation measures.

The four mitigation variables recorded mean scores of:

- Mitigation Option (MO. GHG1) = 4.00
- Mitigation Options (MO. GHG2) = 4.01
- Mitigation Options (MO. GHG3) = 3.77
- Mitigation Options (MO. GHG4) = 3.93

Agreement exceeded 84% for the two highest-rated measures.

However, an important facility-level difference emerged: F (2, 400) = 19.51, $p < .001$, $\eta^2 = .089$.

Facility A: M = 4.07

Facility B: M = 4.06

Facility C: M = 3.65

Tukey HSD demonstrated that Facility C was significantly lower than both Facilities A and B.

This implies that a single manufacturing net-zero strategy may identify common technologies, but implementation readiness or perceived feasibility is not identical.

Facility C therefore requires particular attention to potential:

- technological constraints;
- capital requirements;
- management readiness;
- energy infrastructure;
- implementation barriers; and
- workforce capacity.

Industrial net-zero pathways commonly involve combinations of energy efficiency, material efficiency, electrification, low-carbon energy and fuel switching, circular material flows, and carbon capture for residual process emissions (IPCC, 2022).

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Oil and Gas Mitigation Opportunities

Oil and Gas mitigation-variable means ranged from 3.12 to 3.88.

The strongest mitigation variable recorded:

M = 3.88, agreement = 68.1%.

Importantly, the overall mitigation scores across the three facilities were almost identical:

- Facility A = 3.61
- Facility B = 3.61
- Facility C = 3.63.

ANOVA confirmed: F (2, 404) = .032, p = .969.

Despite differences in respondents' ratings of individual emission-source activities, the three Oil and Gas facilities were statistically indistinguishable in their overall mitigation-option scores.

Therefore, the current evidence supports the development of a common Oil and Gas decarbonization framework, while facility-specific GHG inventories determine precisely where and how aggressively particular measures should be implemented.

Transportation Mitigation Opportunities

Transportation recorded the strongest overall questionnaire support for mitigation measures.

The four-variable means were:

- 4.17;
- 4.24;
- 3.75; and
- 4.00.

The highest-rated option received approximately 91.3% agreement.

Facility mitigation scores were:

- A = 4.06;
- B = 4.07;
- C = 3.99.

ANOVA found no significant difference: F (2, 398) = .868, p = .420.

This provides strong questionnaire-based support for a largely standardized initial net-zero pathway across the sampled transportation facilities.

Solid Waste Mitigation Opportunities

The Solid Waste mitigation variable recorded means ranging from 3.33 to 3.99.

The highest-rated measure recorded:

M = 3.99, with approximately 74.6% agreement.

Facility composite mitigation scores were:

- Facility A = 3.84
- Facility B = 3.62
- Facility C = 3.67.

The facility effect was statistically significant: F (2, 395) = 3.323, p = .037, η^2 = .017.

Tukey HSD identified a significant difference specifically between A and B (p = .038).

Solid Waste requires a more differentiated approach than Transportation. Although common measures can be defined, the level of implementation and facility-specific intervention priorities should be tailored.

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Overall Facility-Level Net-Zero Planning Implications

Table 3: Evidence-Based Net-Zero Planning Classification

Sector	Source Profile	Mitigation Profile	Recommended Planning Approach
Manufacturing	Similar across facilities	Facility C significantly lower	Common pathway + targeted support for C
Oil & Gas	Some facility variation	Highly similar across facilities	Common framework + facility-specific hotspot prioritization
Transportation	Highly similar	Highly similar	Standardized sector pathway feasible
Solid Waste	Broad source similarity	Significant facility variation	Facility-specific implementation pathways

It demonstrates that a net-zero plan should contain common sector-level architecture but allow facility-specific implementation.

Proposed Net-Zero Emissions Plan
Based on the questionnaire evidence, the following five-stage framework is proposed.

Stage 1: Baseline GHG Inventory - Target Setting and Governance
Immediate priority
Every participating facility should first establish a verified physical GHG inventory.
This is where the GHG Protocol should formally enter the implementation process. Its Corporate Standard provides requirements and guidance for corporate-level inventories and is intended to support organizations in developing consistent, transparent emissions information for management and reduction strategies. (GHG Protocol, 2024)

Facility inventories should quantify:
Scope 1: Direct emissions from facility-controlled sources.
Scope 2: Purchased electricity and other acquired energy covered by the reporting boundary.
Scope 3: Material indirect value-chain categories supported by reliable data.

This stage should establish:
Baseline year → activity data → emission factors → tCO_2e inventory → emission hotspots → reduction targets → responsibility → monitoring system.

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Stage 2: Rapid Operational Abatement

Short term

The second stage targets emissions that can potentially be reduced using established operational interventions before major capital replacement.

Priority interventions should be selected according to sector.

Manufacturing

- energy audits;
- energy efficiency;
- optimization of fossil-fuel equipment;
- preventive maintenance;
- reduction of avoidable energy losses.

Oil & Gas

- operational efficiency;
- flare minimization where applicable;
- gas recovery;
- fugitive-emissions monitoring;
- leak detection and repair;
- reduction of avoidable venting.

Transportation

- fleet efficiency;
- route optimization;
- preventive maintenance;
- reduced unnecessary journeys;
- driver fuel-efficiency practices.

Solid Waste

- improved collection;
- reduced uncontrolled disposal/burning;
- segregation;
- recycling/resource recovery;
- route efficiency.

Stage 3: Energy-System Transformation

Once short-term operational improvements have been implemented, facilities should progressively address structural dependence on fossil energy.

The questionnaire demonstrates strong support for mitigation options, particularly in Manufacturing and Transportation.

The transition should emphasize:

- renewable electricity;
- energy-efficient equipment;
- electrification where technically feasible;
- improved electricity-management systems;
- reduced dependence on inefficient fossil-fuel generators;
- cleaner transport technologies; and
- sector-specific lower-carbon energy alternatives.

The IPCC identifies electrification, efficiency, low-carbon energy, fuel switching, and infrastructure transformation among major components of industrial and transport decarbonization (IPCC, 2022).

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Stage 4: Deep Decarbonization

The remaining emissions will likely require larger technological and process changes.

Potential interventions include:

- major equipment replacement;
- deeper process electrification;
- process redesign;
- circular material flows;
- waste-resource recovery;
- low- or zero-GHG energy carriers;
- carbon capture where technically appropriate;
- methane recovery;
- deep fleet transition.

These investments should occur only following robust technical and financial feasibility assessment.

Stage 5: Neutralization of Residual Emissions

The final stage addresses emissions that remain after technically and economically practicable reductions have been implemented.

The sequence should be:

Measure → Avoid → Reduce → Replace → Deeply decarbonize → Neutralize residual emissions

This avoids constructing a net-zero strategy primarily around offsets.

Facility-level residual emissions should first be quantified transparently before credible and durable removal or neutralization mechanisms are considered.

Proposed Implementation Roadmap

Table 4: Facility-Level Net-Zero Roadmap for GHG-Emitting Facilities in Port Harcourt

Stage	Principal Action	Main Outcome
1. Establish	GHG inventory, organizational boundary, targets, MRV	Credible baseline
2. Reduce rapidly	Efficiency, operational controls, fugitive/flare/waste/fleet interventions	Immediate reductions
3. Transform energy	Renewables, electrification, clean energy, fleet transition	Structural reduction in fossil-fuel dependence
4. Deep decarbonization	Process change, technology replacement, circularity, advanced abatement	Reduction of hard-to-abate emissions
5. Neutralize	Address verified residual emissions with credible removals	Net-zero balance

Nigeria's NDC 3.0 includes mitigation actions relating to energy, fuel combustion, electricity generation, transportation, oil and gas, and waste management, providing a national policy context for such facility-level interventions

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Discussion

The most important finding is that the empirical evidence supports neither a completely uniform strategy nor entirely independent facility plans. Instead, the appropriate model is: Sector-wide decarbonization framework and facility-specific implementation.

GHG-Emitting Activities and Net-Zero Priorities in Manufacturing Facilities

The study found broad similarity in respondents' assessment of GHG-emitting activities across the three manufacturing facilities. The composite GHG-source scores were 3.59, 3.63, and 3.61 for Facilities A, B, and C, respectively, and the differences were not statistically significant, F (2, 400) = 0.076, p = .927. This indicates that respondents across the sampled manufacturing facilities generally recognized similar categories of emission-generating activities. The similarity is plausible given the common reliance of manufacturing operations on energy-consuming machinery, production processes, electricity, and fossil-fuel-based energy systems. The Intergovernmental Panel on Climate Change (IPCC, 2022) identifies fuel combustion, process emissions, product use, and waste as important sources of industrial GHG emissions and notes that fuel combustion represents a substantial component of direct industrial emissions. Nigeria's national decarbonization planning similarly identifies industry and energy as major areas requiring coordinated low-carbon interventions (IPCC, 2022).

However, a substantially different pattern emerged for mitigation opportunities. Although the manufacturing sector generally demonstrated strong support for mitigation measures, significant differences were observed between facilities, F (2, 400) = 19.51, p < .001, η^2 = .089. Facilities A (M = 4.07) and B (M = 4.06) recorded significantly higher mitigation-option scores than Facility C (M = 3.65). Thus, facilities may face similar GHG-generating activities without necessarily possessing identical perceptions, preparedness, or opportunities for mitigation.

This finding has important implications for facility-level net-zero planning. Industrial decarbonization is rarely dependent on a single technology. The IPCC identifies energy efficiency, material efficiency, electrification, low- or zero-GHG energy carriers, circular-economy approaches, process changes, and carbon capture as complementary industrial mitigation pathways. Nigeria's Energy Transition Plan likewise envisages major changes in electricity generation and industrial energy systems as components of the country's transition towards net zero (IPCC, 2022).

Consequently, the lower mitigation score observed in Manufacturing Facility C should not be interpreted as evidence that the facility has no reduction potential. Instead, it indicates that facility-specific conditions may influence the perceived applicability or readiness for mitigation measures. Such conditions could include differences in technologies, capital requirements, existing equipment, energy supply systems, management priorities, and organizational capacity. These factors were not directly measured in the present study and should therefore be investigated during the detailed facility-level feasibility stage. The finding supports the proposed net-zero approach of developing a common manufacturing decarbonization framework while allowing facility-specific implementation pathways.

Oil and Gas Facilities: Similar Mitigation Profiles but different assessments of GHG-Emitting Activities

The Oil and Gas findings are particularly important because facilities operating within the same sector would generally be expected to exhibit similar categories of GHG-emitting activities and mitigation opportunities. However, the analysis revealed that while respondents identified broadly comparable mitigation opportunities across the facilities, the relative importance assigned to specific GHG-emitting activities differed significantly among the facilities. The present findings do not fully support that expectation. Having the same category does not mean having the same emission profile or mitigation potential.

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The three facilities differed significantly in their composite assessment of GHG-emitting sources, F (2, 404) = 5.62, p = .004, η^2 = .027. Facility A recorded a mean of 3.73 compared with 3.41 for Facility B and 3.57 for Facility C. However, this result should not be interpreted to mean that the facilities necessarily have entirely different categories of emission sources. Instead, the relative importance attributed to those emission-generating activities differed among respondents.

This distinction is consistent with literature on Nigeria's oil and gas sector. Raimi et al. (2025) identify gas flaring and fugitive releases among activities associated with methane emissions in the Niger Delta and highlight limitations in monitoring and reporting that complicate accurate facility-level emissions assessment. Abdulsamad and George-Ikoli (2024) similarly identify flaring, fugitive emissions, and venting as important methane-release pathways within Nigeria's oil and gas value chain and argue for facility-appropriate monitoring, reporting, verification, and leak-detection systems.

Therefore, multiple oil and gas facilities can possess similar categories of emission-generating activities while differing in the relative significance of those activities. Differences in operational configuration, production characteristics, equipment, flaring practices, venting, and fugitive-emission control may contribute to differing source profiles. These operational explanations should, however, be confirmed through physical facility inventories rather than inferred solely from questionnaire responses.

The mitigation findings are perhaps even more important. Despite the variation in GHG-source scores, mitigation-option scores were almost identical across Oil and Gas Facilities A (M = 3.61), B (M = 3.61), and C (M = 3.63), and the difference was not statistically significant, F (2, 404) = .032, p = .969.

Thus, the data shows very strong similarity in overall mitigation perceptions across the three Oil and Gas facilities. This result agrees with current Nigerian methane-abatement literature. Recent work identifies leak detection and repair, vapour recovery, replacement of gas-driven pneumatic equipment, and improved flaring practices as important technical abatement approaches for Nigeria's oil and gas industry. Nigerian methane-management research similarly emphasizes flare-gas capture, improved monitoring, methane-management systems, and operational interventions as important mitigation strategies.

The implication for the proposed net-zero plan is therefore clear: Oil and Gas facilities can operate under a broadly common mitigation framework, but facility-specific GHG inventories should determine which emission pathway deserves the greatest attention at each facility.

Transportation Facilities and the Potential for a Common Net-Zero Transition Framework

Transportation produced the greatest level of statistical consistency among the sampled facilities. No significant differences occurred for GHG-source scores, F (2, 398) = .074, p = .928, or mitigation-option scores, F (2, 398) = .868, p = .420. Mitigation scores were high and closely clustered across Facilities A (M = 4.06), B (M = 4.07), and C (M = 3.99). The similarity is consistent with the broader characteristics of Nigeria's transportation sector. Akujor et al. (2022) reported extensive dependence on fossil fuels across Nigerian transport subsectors and proposed decarbonization through measures such as public transport, rail and water transport revitalization, low-carbon fuels, and electric vehicles. The relatively uniform mitigation scores also align with Nigeria's Energy Transition Plan, which proposes transport decarbonization through lower-emission technologies, mode shifting, biofuels as an interim measure, and progressive electrification. More recent Nigerian research supports electric mobility as a potentially important emissions-reduction pathway but identifies electricity reliability, vehicle costs, import duties, and charging infrastructure as material implementation barriers. The present findings therefore support the development of a relatively standardized initial net-zero framework for the sampled transportation facilities. Actions such as fleet-efficiency improvement, vehicle maintenance, journey and route optimization, cleaner fuels, modal change, and eventual electrification could form common elements of such a framework.

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However, statistical similarity should not be interpreted as meaning that the three transportation facilities have identical physical emissions. Their actual fuel consumption, fleet size, vehicle age, utilization, journey distances, and operational patterns could differ. Consequently, common mitigation measures should be accompanied by facility-specific baselines and targets.

Solid Waste Facilities and the Need for Facility-Specific Decarbonization

Solid Waste showed greater facility-level variability than Transportation and Oil and Gas, particularly regarding mitigation options. While the facilities did not differ significantly in mean GHG-source scores, F (2, 395) = .509, p = .601, mitigation scores differed significantly, F (2, 395) = 3.323, p = .037, η^2 = .017. Facility A recorded the highest mitigation score (M = 3.84), followed by Facility C (M = 3.67) and Facility B (M = 3.62). Furthermore, the earlier correlation analysis revealed several significant relationships within Solid Waste. For example, health and safety impacts were positively associated with mitigation options in Facility A (r = .319, p < .001), while GHG-type recognition was positively related to mitigation options in Facility B (r = .332, p < .001). Those relationships indicate greater interconnection among respondents' perceptions of GHG characteristics, impacts, and mitigation within the Solid Waste facilities, although they should not be interpreted as causal.

The observed variability is consistent with the characteristics of municipal waste systems. Methane generation from municipal solid waste is strongly influenced by waste composition, decomposition, and disposal conditions, making the waste sector an important target for methane mitigation.

The Nigerian evidence is particularly relevant. Ayandele et al. (2024), in their methane-mitigation work for Lagos waste management, emphasized the need to connect waste methane information with practical decision-making tools and interventions. Nigeria's current waste-sector policy direction also seeks improved solid waste-management systems and recognizes existing gaps in waste-management practice and infrastructure. These studies help explain why a single mitigation technology may not be appropriate for every solid-waste facility. The optimal intervention may depend on factors such as waste composition, collection arrangements, segregation, recycling, treatment, landfill management, methane recovery potential, and transport systems.

Accordingly, the Solid Waste result strongly supports a facility-level hierarchy of interventions rather than uniform implementation. Waste prevention and reduction, separation, recycling and material recovery, improved collection systems, organic waste treatment, engineered disposal, and landfill-gas management should be evaluated according to the specific conditions at each facility.

Implications of the Findings for the Proposed Net-Zero Plan

Taken together, the findings highlight an important distinction between sector-wide strategy and facility-level implementation. The results do not support the assumption that every facility requires a completely independent net-zero framework. Neither do they support applying exactly the same interventions to all facilities merely because they belong to the same sector.

Instead, the findings suggest the following model:

National net-zero ambition → Sector decarbonization pathway → Facility GHG inventory → Facility-specific intervention priorities → Implementation → Monitoring and verification → Progressive reduction of residual emissions.

This approach is consistent with Nigeria's broader climate-policy direction. Nigeria's Energy Transition Plan identifies power, cooking, oil and gas, transportation, and industry as key sectors in its pathway towards the national 2060 net-zero objective.

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The country's broader deep-decarbonization research likewise examines different pathways across sectors rather than assuming that one technology or intervention will deliver equivalent results everywhere.
The study's findings provide empirical support for this differentiated approach. Manufacturing requires a shared decarbonization architecture but with additional attention to Facility C's mitigation circumstances. Oil and Gas can utilize a broadly common mitigation framework while applying facility-level hotspot analysis to determine the relative importance of flaring, combustion, venting, fugitive emissions, and other relevant activities. Transportation shows sufficient similarity to justify a more standardized initial approach. Solid Waste requires greater facility-specific adaptation because mitigation perceptions and GHG-related response patterns vary more substantially between facilities.

The finding that sector-specific pathways are necessary is consistent with emerging Nigerian decarbonization research. Scenario modelling of Nigerian industrial subsectors has shown that different industries can respond differently to the same decarbonization assumptions, supporting differentiated technological pathways rather than a uniform industrial policy.

From Research Finding to a Credible Net-Zero Implementation Framework
An important methodological implication of this study is that the questionnaire data should be viewed as a prioritization and planning tool, rather than a substitute for a physical GHG inventory.
The Greenhouse Gas Protocol Corporate Standard is designed to enable organizations to prepare consistent corporate-level GHG inventories and generate information useful for GHG management and reduction strategies. Therefore, the first implementation step arising from this study should be the establishment of verified facility baselines, followed by identification of material emissions sources and development of measurable reduction targets. (GHG Protocol, 2024).
The questionnaire results can then inform where those detailed assessments should focus. For example, strong recognition of particular mitigation measures can guide feasibility studies and stakeholder engagement, while statistically significant facility differences can identify where a uniform intervention may require local adaptation. This distinction strengthens rather than weakening the proposed net-zero plan. It means the study provides evidence concerning what respondents and facilities identify as relevant emission challenges and mitigation opportunities, while subsequent GHG inventories provide the physical tCO_2e values required to establish reduction trajectories.

Manufacturing illustrates this particularly clearly. GHG-source assessments were remarkably similar across Facilities A, B, and C, suggesting common underlying operational challenges. However, mitigation-option scores differed substantially, with Facility C reporting significantly lower scores. Therefore, identical source characteristics do not necessarily imply identical mitigation readiness.
This finding is compatible with the wider industrial decarbonization literature. Industrial net zero normally requires combinations of efficiency, electrification, low-carbon energy, material efficiency, circular economy strategies, and technologies targeting residual process emissions. The applicability and timing of these interventions vary depending on existing plant and infrastructure (IPCC, 2022).

Oil and Gas presents a different finding. Source assessments showed some variation between facilities, but mitigation-option scores were virtually identical. This is particularly relevant to the concern that facilities within the same petroleum sector would be expected to exhibit similar GHG sources and mitigation opportunities. The current study provides partial support for that expectation. Common source categories may indeed occur, while their relative prominence varies according to facility operations. At the same time, the absence of significant differences in mitigation scores indicates broad commonality in the perceived mitigation framework.

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Transportation demonstrated the greatest consistency. Neither source nor mitigation scores differed significantly across facilities. Accordingly, common interventions can reasonably form the initial basis of a transport-facility net-zero framework, although future physical inventories should determine facility-specific reduction targets. This finding is compatible with the broader requirement for systemic transport transformation involving cleaner technologies, efficiency, demand management, electrification, and infrastructure changes (IPCC, 2022).

Solid Waste exhibited greater mitigation heterogeneity. This implies that the same waste-sector intervention should not automatically be applied at identical intensity across different facilities. Differences in waste streams, collection systems, infrastructure, treatment methods, existing recovery systems, equipment, and technological readiness may affect which interventions are appropriate.

Taken together, the results demonstrate why moving directly from a national net-zero target to a single facility strategy would be inappropriate. Nigeria's NDC 3.0 itself adopts differentiated mitigation measures across sectoral emission categories and sets a pathway towards net zero by 2060. Facility-level planning should follow the same principle of differentiated intervention.

Conclusion

This study used questionnaire evidence from 12 facilities across Manufacturing, Oil and Gas, Transportation, and Solid Waste sectors to develop an evidence-informed framework for facility-level net-zero planning in Port Harcourt. The results demonstrate that respondents identify substantial GHG-emitting activities and mitigation opportunities across all four sectors, but the degree of facility-level uniformity differs between sectors.

Manufacturing facilities demonstrated similar GHG-source profiles but significantly different mitigation scores, with Facility C requiring particular attention. Oil and Gas facilities showed some differences in source assessments but almost identical overall mitigation scores. Transportation demonstrated substantial consistency across both emission-source and mitigation dimensions. Solid Waste showed greater facility-level variability in mitigation opportunities.

These findings demonstrate that achieving facility-level net zero requires a combination of common sector-level decarbonization architecture and facility-specific implementation plans.

The proposed five-stage pathway begins with verified GHG accounting and governance, followed by rapid operational abatement, energy-system transformation, deep technological decarbonization, and eventual neutralization of verified residual emissions. Most importantly, the questionnaire evidence should be used to prioritize where facility investigations and interventions should begin, rather than being interpreted as a substitute for a physical emissions inventory.

Accordingly, the next practical step for GHG-emitting facilities in Port Harcourt is to translate the priorities identified in this study into inventory-based facility plans to contain quantified Scope 1, Scope 2, and relevant Scope 3 emissions, time-bound reduction targets, capital requirements, responsibilities, key performance indicators, and monitoring, reporting and verification arrangements. The GHG Protocol provides an established framework for corporate-level inventory development, while Nigeria's NDC 3.0 provides the national policy direction towards net-zero emissions by 2060 (GHG Protocol, 2024).

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